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TECHNOLOGICAL UNIVERSITY DUBLIN
CITY CAMPUS - GRANGEGORMAN

TU856 - BSc (Hons) Computer Science

Year 3

SEMESTER 1 EXAMINATIONS 2024/25

Mobile Software Development

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Exam Duration: 2 hours

Instructions:

Answer ALL Questions

Write your exam number at the top of the paper

All questions should be answered in the exam paper

Section 1: MULTIPLE CHOICE QUESTIONS

Answer the following questions on the exam paper. All multiple choice questions are worth 1.5 marks for a correct answer, $-\frac{1}{2}$ for an incorrect answer and 0 for no answer. In total, the questions in this section are worth 60 marks. Circle the letter A, B, C or D.

1. Which of the following best describes a native mobile app?

- A. An app that runs on any operating system without installation
- B. An app developed for a specific operating system
- C. A website optimized for mobile devices
- D. An app that does not require an internet connection

(1.5 marks)

2. Which of the following is NOT a characteristic of a native app?

- A. Runs independently of a web browser
- B. Can access device hardware, like GPS and camera
- C. Needs a specific operating system to function
- D. Built using HTML, CSS, and JavaScript only

(1.5 marks)

3. What sequence of methods is called when an activity is first started?

- A. onCreate() -> onResume() -> onStart()
- B. onCreate() -> onStop() -> onDestroy()
- C. onCreate() -> onStart() -> onResume()
- D. onPause() -> onResume() -> onStop()

(1.5 marks)

4. If an activity is in the background and the user navigates back to it, which lifecycle method will be called first for this activity?

- A. onDestroy()
- B. onPause()
- C. onStart()
- D. onResume()

(1.5 marks)

5. Which lifecycle methods are always called, even if an activity is closed unexpectedly?

- A. onPause() and onResume()
- B. onPause() and onStop()
- C. onResume() and onCreate()
- D. onStop() and onDestroy()

(1.5 marks)

6. Which XML attribute is used to set the width of a view to fill the entire width of the parent's view?

- A. wrap_content
- B. layout_alignParentLeft
- C. layout_centerInParent
- D. match_parent

(1.5 marks)

7. If you want a view to be centered horizontally in a ConstraintLayout, which constraints should you set?

- A. layout_constraintBottom_toTopOf and layout_constraintTop_toBottomOf
- B. layout_alignParentCenter
- C. layout_constraintLeft_toLeftOf and layout_constraintRight_toRightOf
- D. layout_alignParentLeft and layout_alignParentRight

(1.5 marks)

8. Which of the following layouts does NOT allow direct alignment of child views relative to one another?

- A. ConstraintLayout
- B. RelativeLayout
- C. TableLayout
- D. LinearLayout

(1.5 marks)

9. What is the main advantage of using ConstraintLayout over RelativeLayout in terms of performance?

- A. ConstraintLayout allows overlapping views
- B. ConstraintLayout uses less XML code
- C. ConstraintLayout reduces nested view hierarchies
- D. ConstraintLayout allows scrolling by default

(1.5 marks)

10. What is an Android activity?

- A. A background service that runs when the app is closed
- B. A single screen with a user interface
- C. A type of broadcast receiver
- D. A thread for handling tasks

(1.5 marks)

11. Which of the following best describes an intent in Android?

- A. A graphical component in Android UI
- B. A message object
- C. A layout used to create complex designs
- D. A data storage object

(1.5 marks)

12. Which method is used to pass data between activities in an intent?

- A. putExtra()
- B. startActivityForResult()
- C. getData()
- D. setIntent()

(1.5 marks)

13. Which of the following is an example of an implicit intent?

- A. Starting a background service within the app
- B. Requesting an activity to send an email
- C. Calling finish() to close an activity
- D. Opening a specific activity within an app

(1.5 marks)

14. What is the primary purpose of an intent filter in Android manifest file?

- A. To determine how data is saved in the app
- B. To specify the types of intents it can handle
- C. To control the flow of data between activities
- D. To increase app performance

(1.5 marks)

15. Which method retrieves the data from an intent in the receiving activity?

- A. setExtras()
- B. getExtras()
- C. getIntent().getData()
- D. getIntent().getExtras()

(1.5 marks)

16. What will happen if an explicit intent is used to start an activity that is not declared in the Android manifest file?

- A. The intent will be redirected to the main activity
- B. The app will ignore the intent and continue running
- C. The app will crash
- D. The app will display a blank screen with return button

(1.5 marks)

17. Which of the following is true about Intent.ACTION_VIEW?

- A. It opens a specified activity in the same app
- B. It displays data to the user using the most appropriate app
- C. It starts a background service to download data
- D. It retrieves data from the clipboard

(1.5 marks)

18. What is ListView in Android?

- A. A view that manages fragment transactions
- B. A button with multiple states
- C. A view that displays a vertically scrolling list of items
- D. A view that displays images in a list

(1.5 marks)

19. Which method in a custom view should be overridden to handle the drawing of a custom view's content?

- A. onMeasure()
- B. onCreate()
- C. onLayout()
- D. onDraw()

(1.5 marks)

20. In Android, which method is used to get the size (exact width and height) of a view?

- A. measure()
- B. onDraw()
- C. layout()
- D. getSize()

(1.5 marks)

21. How can you create a custom view in Android?

- A. Override the onCreate() method in MainActivity
- B. Use a RelativeLayout with custom attributes
- C. Extend the View or a subclass of View and override onDraw()
- D. Extend the Activity class and override onDraw()

(1.5 marks)

22. Which of the following files is used to define themes in an Android app?

- A. strings.xml
- B. styles.xml
- C. dimens.xml
- D. colors.xml

(1.5 marks)

23. How would you retrieve a string resource programmatically in an activity?

- A. getStringResource(R.string.resourceName)
- B. findViewById(R.string.resourceName)
- C. getResources().getString(R.string.resourceName)
- D. getText(R.string.resourceName)

(1.5 marks)

24. What is the purpose of dimens.xml in Android resources?

- A. To specify colors used in the app
- B. To define values like padding, margin, and font sizes
- C. To define image resources
- D. To store string values like name

(1.5 marks)

25. If you want to specify a value only for devices with Android API level 24 or higher, which directory name would you use?

- A. drawable-mdpi
- B. layout-24
- C. values
- D. values-v24

(1.5 marks)

26. How would you specify a unique string for German-speaking users in Android resources?

- A. res/strings-de
- B. res/de-strings
- C. res/values
- D. res/values-de

(1.5 marks)

27. Which of the following measures to be used with views

- A. Actual Pixels (px)
- B. Scalable Pixels (sp)
- C. Actual Measurement (in, mm)
- D. Density-independent Pixels (dp)

(1.5 marks)

28. In a Room database, what does DAO stand for?

- A. Data Annotation Object
- B. Database Access Object
- C. Data Analytics Object
- D. Data Access Object

(1.5 marks)

29. What is the purpose of a Service in Android?

- A. To perform short-running operations in the background
- B. To manage network connections in the background
- C. To perform long-running operations in the background
- D. To handle UI interactions

(1.5 marks)

30. Which method is used to stop a Service after it's been started with startService()

- A. unbindService()
- B. terminateService()
- C. closeService()
- D. stopSelf()

(1.5 marks)

31. Which Android component is best suited for scheduling recurring tasks that need to execute at precise intervals, even after the device reboots?

- A. WorkManager
- B. AsyncTask
- C. AlarmManager
- D. JobScheduler

(1.5 marks)

32. What is the Fused Location Provider API in Android?

- A. A method to combine Wi-Fi and Bluetooth signals for location
- B. An alternative to GPS and network providers
- C. A database of location data from various apps
- D. A method that combines multiple location sources to simplify location data

(1.5 marks)

33. Which of the following is an Android best practice for requesting location updates to conserve battery life?

- A. Use multiple location providers simultaneously
- B. Use `requestSingleUpdate()` when you only need the user's current location once
- C. Continuously request location updates in the foreground
- D. Request updates as frequently as possible, so you always have the location information

(1.5 marks)

34. Which of the following methods allows an Android app to check if location services are enabled on the device?

- A. `verifyLocationProvider()`
- B. `checkLocationSettings()`
- C. `hasLocationPermissions()`
- D. `isLocationEnabled()`

(1.5 marks)

35. When implementing location updates in an Android app, which lifecycle method is most appropriate for starting location updates?

- A. `onCreate()`
- B. `onPause()`
- C. `onStart()`
- D. `onResume()`

(1.5 marks)

36. Which of the following techniques helps improve battery performance in Android?
- A. Minimizing background processes
 - B. Polling the network frequently for updates
 - C. Running intensive processes in the foreground
 - D. Using onStart() to trigger heavy network requests
- (1.5 marks)
37. Which Android tool allows developers to monitor app performance, memory usage, and network usage?
- A. Profiler
 - B. Performance Manager
 - C. Logcat
 - D. Android Monitor
- (1.5 marks)
38. To optimize rendering performance in Android, what should be avoided in complex UIs?
- A. ConstraintLayout
 - B. Fragments
 - C. Activities
 - D. Nested layouts
- (1.5 marks)
39. What is the primary purpose of using RecyclerView instead of ListView?
- A. To enhance the use of multiple views within each item
 - B. To provide better scrolling performance for dynamic lists
 - C. To avoid using adapters
 - D. To support drag and drop
- (1.5 marks)
40. What is the primary purpose of using MVC in Android app development?
- A. To increase battery efficiency
 - B. To achieve a clear separation of concerns between UI, data, and business logic
 - C. To simplify the app's lifecycle management
 - D. To reduce app size
- (1.5 marks)

Section 2: LONG QUESTIONS

Answer the following questions on the exam paper. In total, the questions in this section are worth 40 marks.

1. With reference to activities and fragments, Why are fragments used? Could we create a Multiscreen application using multiple activities instead?

(10 marks)

2. With the aid of an example, explain what a Back Stack is, and why it is important for your application?

(10 marks)

3. Android imposes a couple of rules to maintain the responsiveness of the device. List these rules and explain why it is important to determine when an application has become unresponsive. As a developer, what should you do to observe these rules?

(10 marks)

4. An Android developer has set a minTime of 3 milliseconds, and a minDistance of 2 metres. Explain how each of these two parameters impact the triggering of the onLocationChanged event handler.

(10 marks)

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End of Exam

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